Econometrics Assignment 1

OLS Regression With Real-World Data

Assignment Overview

This assignment introduces applied Ordinary Least Squares (OLS) regression using real-world datasets.

The assignment focuses on:

- simple linear regression
- multiple regression
- omitted variable bias
- controls and dummy variables
- interaction terms
- log-linear models
- inference in regression
- heteroskedasticity
- multicollinearity
- residual analysis
- robust standard errors

You are expected to approach the assignment as both:

- a statistical exercise
- an economic/business analysis problem

You must use `statsmodels` for estimation and inference. Do not rely only on `sklearn`.

Topics to Learn Before Attempting This Assignment

A. Simple Linear Regression

- OLS intuition
- line fitting
- residuals
- SSE and MSE
- interpretation of coefficients
- R^2

B. Multiple Regression

- omitted variable bias
- controls
- dummy variables
- interaction terms
- nonlinear transformations
- log models

C. Inference in Regression

- t-tests
- F-tests
- ANOVA
- ANCOVA
- confidence intervals in regression
- hypothesis testing
- variance decomposition

D. OLS Assumptions and Diagnostics

- exogeneity
- linearity
- homoskedasticity
- heteroskedasticity
- multicollinearity
- autocorrelation
- independence
- endogeneity
- robust standard errors

Required Submission

Submit:

1. One Jupyter notebook
2. One short report

Your report must include:

- research question
- dataset description
- regression equations
- coefficient interpretation
- t-tests and F-tests
- robust standard errors
- residual analysis
- heteroskedasticity diagnostics
- multicollinearity diagnostics
- final economic or business interpretation

Problem 1 — Fuel Efficiency and Car Design

Dataset

Use the **Auto MPG Dataset** from UCI.

The dataset contains:

- miles per gallon (MPG)
- cylinders
- displacement
- horsepower
- weight
- acceleration
- model year
- origin

Some observations with missing values were removed in the cleaned dataset.

Research Question

What car features explain fuel efficiency, and how much does vehicle weight matter after controlling for engine and model characteristics?

Variables

Dependent Variable

- mpg

Main Explanatory Variable

- weight

Controls

- horsepower
- cylinders
- displacement
- acceleration
- model_year
- origin

Part A — Simple Regression

Estimate:

$$mpg = \beta_0 + \beta_1 weight + u$$

Tasks

1. Estimate the regression model.
2. Plot:
    - fitted regression line
    - residuals
3. Report:
    - SSE
    - MSE
    - R^2

Questions

1. What is the sign of β_1?
2. Interpret β_1 in plain English.
3. Is the estimated relationship economically reasonable?

Expected intuition: heavier cars should have lower MPG.

Part B — Multiple Regression

Estimate:

$$mpg = \beta_0 + \beta_1 weight + \beta_2 horsepower + \beta_3 displacement + \beta_4 cylinders + u$$

Then extend the model by adding:

- model year
- origin dummy variables

Tasks

1. Estimate both models.
2. Compare coefficients across models.
3. Interpret origin dummy variables carefully.

Questions

1. How does the coefficient on weight change after adding controls?
2. Was the simple regression likely suffering from omitted variable bias?
3. Are newer model-year cars more fuel efficient after controlling for other variables?
4. What do the origin dummy variables imply?

Part C — Nonlinear Model

Estimate:

$$\log(mpg) = \beta_0 + \beta_1 \log(weight) + \beta_2 \log(horsepower) + controls + u$$

Tasks

1. Estimate the log-log model.
2. Compare residual plots across models.
3. Compare R^2 values.

Questions

1. Interpret β_1 as an elasticity.
2. Does the log transformation improve residual behavior?
3. Which specification appears more appropriate?

Part D — Inference and Diagnostics

Tasks

Perform the following:

- t-test for weight
- F-test for joint significance of:
  - horsepower
  - displacement
  - cylinders
- robust standard errors
- heteroskedasticity test
- VIF analysis for multicollinearity

Questions

1. Is weight statistically significant?
2. Are engine-related variables jointly significant?
3. Is there evidence of heteroskedasticity?
4. Is multicollinearity a concern?

Conclusion

Write a short conclusion:

> "After controlling for engine size and model year, vehicle weight has / does not have an independent association with MPG."

Problem 2 — Housing Prices and Location Effects

Dataset

Use the **King County House Sales Dataset** from Kaggle.

The dataset contains house-sale information for homes sold between May 2014 and May 2015.

Research Question

How much do size, condition, renovation, and location explain housing prices?

This problem is intended to develop intuition for:

- omitted variable bias
- location controls
- log-price models
- fixed effects

Variables

Dependent Variable

- price

Main Variables

- sqft_living
- bedrooms
- bathrooms
- floors
- condition
- grade
- yr_built

- yr_renovated
- waterfront
- view
- zipcode
- lat
- long

Part A — Baseline Model

Estimate:

$$price = \beta_0 + \beta_1 sqft\_living + u$$

Tasks

1. Estimate the model.
2. Plot:
   - price vs sqft_living
3. Check whether the relationship appears linear.

Questions

1. Interpret β_1.
2. Is the interpretation economically realistic?
3. Does the relationship appear linear?

Part B — Multiple Regression With Controls

Estimate:

$$
\begin{aligned}
price = \beta_0 &+ \beta_1 sqft\_living \\
&+ \beta_2 bedrooms \\
&+ \beta_3 bathrooms \\
&+ \beta_4 floors \\
&+ \beta_5 condition \\
&+ \beta_6 grade \\
&+ \beta_7 waterfront \\
&+ \beta_8 view \\
&+ u
\end{aligned}
$$

Questions

1. What happens to the coefficient on sqft_living after controls?
2. Why might the simple regression overstate or understate the size effect?
3. Interpret waterfront.
4. Interpret grade.

Part C — Log Model

Estimate:

$$\begin{aligned}\log(price) = \beta_0 &+ \beta_1 \log(sqft\_living) \\ &+ \beta_2 bedrooms \\ &+ \beta_3 bathrooms \\ &+ \beta_4 condition \\ &+ \beta_5 grade \\ &+ \beta_6 waterfront \\ &+ \beta_7 view \\ &+ u\end{aligned}$$

Questions

1. Interpret β_1.
2. Interpret waterfront.
3. Which model appears better:
   - level-price model
   - log-price model
4. Use residual plots to justify your answer.

Part D — Dummy Variables and Location Controls

Add zipcode fixed effects:

$$
\begin{aligned}
\log(price) = \beta_0 &+ \beta_1 \log(sqft\_living) \\
&+ \beta_2 bedrooms \\
&+ \beta_3 bathrooms \\
&+ \beta_4 condition \\
&+ \beta_5 grade \\
&+ zipcode\ dummies \\
&+ u
\end{aligned}
$$

Questions

1. What happens to β_1 after adding zipcode controls?
2. Why is location an omitted variable risk?
3. Are zipcode dummies jointly significant?
4. Does adding location improve R^2?

Part E — Interaction Term

Estimate:

$$\begin{aligned}\log(price) = \beta_0 &+ \beta_1 \log(sqft\_living) \\ &+ \beta_2 waterfront \\ &+ \beta_3 \left[\log(sqft\_living) \times waterfront\right] \\ &+ controls \\ &+ u\end{aligned}$$

Questions

1. Does size matter differently for waterfront houses?
2. Interpret the interaction term.
3. Is the interaction statistically significant?

Part F — Diagnostics

Tasks

Perform:

- robust standard errors
- Breusch-Pagan test
- VIF analysis
- residual plots

Questions

1. Are OLS assumptions plausible here?
2. Is heteroskedasticity present?
3. Is multicollinearity severe?
4. Why should the conclusions be interpreted as associations rather than causal effects?

Problem 3 — Marketing Response and Customer Spending

Dataset

Use the **Customer Personality Analysis / Marketing Campaign Dataset** from Kaggle.

The dataset contains:

- demographics
- campaign responses
- spending behavior
- purchase channels

Business Question

Which customer characteristics are associated with higher spending, and which customers are more likely to respond to campaigns?

Approach this problem like a business analyst advising a marketing team.

Construct

Create:

$$\begin{aligned} total\_spending = & MntWines + MntFruits \\ & + MntMeatProducts \\ & + MntFishProducts \\ & + MntSweetProducts \\ & + MntGoldProds \end{aligned}$$

Possible Explanatory Variables

- Income
- Kidhome

- Teenhome
- Age
- Education
- Marital_Status
- Recency
- NumWebPurchases
- NumCatalogPurchases
- NumStorePurchases
- AcceptedCmp1–5
- Response

Part A — Spending Model

Estimate:

$$total\_spending = \beta_0 + \beta_1 Income + u$$

Tasks

1. Estimate the model.
2. Plot:
   - spending vs income
3. Check for outliers.

Questions

1. Interpret β_1.
2. Is the relationship economically meaningful?
3. Does income alone explain spending well?

Part B — Multiple Regression With Controls

Estimate:

$$
\begin{aligned}
total\_spending = \beta_0 &+ \beta_1 Income \\
&+ \beta_2 Age \\
&+ \beta_3 Kidhome \\
&+ \beta_4 Teenhome \\
&+ \beta_5 Recency \\
&+ \beta_6 NumWebPurchases \\
&+ \beta_7 NumCatalogPurchases \\
&+ \beta_8 NumStorePurchases \\
&+ u
\end{aligned}
$$

Questions

1. Does income still matter after adding controls?
2. Interpret Kidhome and Teenhome.
3. Which purchase channel is most strongly associated with spending?
4. Are purchase-channel variables jointly significant?

Part C — Log Spending Model

Create:

$$log\_total\_spending = \log(total\_spending + 1)$$

$$log\_income = \log(Income)$$

Estimate:

$$\begin{aligned} log\_total\_spending = \beta_0 &+ \beta_1 log\_income \\ &+ \beta_2 Age \\ &+ \beta_3 Kidhome \\ &+ \beta_4 Teenhome \\ &+ \beta_5 Recency \\ &+ controls \\ &+ u \end{aligned}$$

Questions

1. Interpret β_1 as an elasticity.
2. Why do we use $\log(total\_spending + 1)$?
3. Does the log model reduce heteroskedasticity?

Part D — Linear Probability Model

Create binary dependent variable:

$$Response$$

Estimate:

$$\begin{aligned} Response = \beta_0 &+ \beta_1 Income \\ &+ \beta_2 Age \\ &+ \beta_3 Kidhome \\ &+ \beta_4 Teenhome \\ &+ \beta_5 Recency \\ &+ \beta_6 total\_spending \\ &+ u \end{aligned}$$

Even though the dependent variable is binary, use OLS as a Linear Probability Model.

Questions

1. Interpret coefficients as probability changes.
2. Which variables are associated with campaign response?
3. Do predicted probabilities fall below 0 or above 1?
4. Why might logistic regression be more appropriate?

Use robust standard errors.

Part E — Business Recommendation Memo

Write a short memo answering:

> "Based on the regression results, which customers should the company target for the next campaign?"

Your memo must include:

- high-value customer profile
- whether income alone is sufficient for targeting
- whether past purchasing behavior matters
- limitations of the analysis
- one better experiment or A/B test the company could run

General Guidelines

- Clearly label every regression.
- Always interpret coefficients economically, not only statistically.
- Use robust standard errors whenever appropriate.
- Include plots wherever useful.
- Discuss assumptions before making strong conclusions.
- Distinguish carefully between:
  - association
  - causation

Main Learning Goals

By the end of this assignment, you should understand:

- how OLS regression works
- how controls affect coefficients
- omitted variable bias
- interpretation of regression outputs
- statistical inference in regression
- regression diagnostics
- heteroskedasticity and robust standard errors
- multicollinearity
- limitations of observational analysis